Growth chains for near-extremal directionally convex polyhexes

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Abstract

This note records a proof, on a class wider than the edge-perimeter minimizers, of the one-cell growth theorem of the author's paper *Shortest-side skeletons of edge-perimeter-minimizing polyhexes*: every directionally convex polyhex S with $|S| \geq M(\sigma(S)-1)$, where $M(t) = \lfloor (t^2+3)/12 \rfloor$ and σ is the edge semiperimeter, admits a one-cell growth sequence with edge-perimeter excess at most two at every stage. The main paper needs the statement only for minimizers and proves it there by a shorter route; the argument here closes the induction on the wider class by a finite case analysis of the tight hull. The common framework of the main paper is reproduced for self-containment.

1 The hexagonal lattice, runs, and directional convexity

This part reproduces, for self-containment, the common framework of the main paper [Utsunomiya, *Shortest-side skeletons of edge-perimeter-minimizing polyhexes*]: the lattice, the perimeter functional, directional convexity, the tight hull and its identities, and the extremal capacity function. Numbering follows this note.

1.1 Lattice and polyhexes

Cells of the hexagonal tiling are indexed by axial coordinates $(q, r) \in \mathbb{Z}^2$. We call q, r and

$$s := q + r$$

the three *keys* of a cell; in words, the third key of a cell is the sum of its two axial coordinates. Two cells are *adjacent* when they differ by one of the six vectors

$$d_0 = (1,0), \quad d_1 = (0,1), \quad d_2 = (-1,1), \quad d_3 = (-1,0), \quad d_4 = (0,-1), \quad d_5 = (1,-1).$$

A *polyhex* is a finite nonempty set of cells that is connected with respect to adjacency. We write $|S|$ for the number of cells of S.

The following four facts are immediate from the definitions and are used throughout without further comment.

Fact 1.1. Two distinct cells are adjacent if and only if they agree in exactly one key and differ by exactly 1 in the other two. In words: an adjacency step keeps one of the three keys fixed and changes each of the other two by one unit (the step $d_0 = (1,0)$, for instance, sends (q, r, s) to $(q+1, r, s+1)$, so the two changing keys need not move in opposite senses).

Proof. Each d_j changes (q, r, s) by $(1,0,1)$, $(0,1,1)$, $(-1,1,0)$, $(-1,0,-1)$, $(0,-1,-1)$ or $(1,-1,0)$, and these are exactly the six triples with one zero entry and two entries equal to ± 1 that are realized. Conversely a difference of that shape is one of the d_j. □

Fact 1.2. A *line* is a set of cells on which one key is constant. Two lines belonging to different keys meet in exactly one cell; two distinct lines belonging to the same key are disjoint.

Fact 1.3. One adjacency step changes each of q, r and s by at most 1.

Fact 1.4. For a polyhex S and each key, the set of values taken by that key on S is an interval of consecutive integers. In words: no key value inside the range of S can be skipped.

Proof. Let k be a key and let $x, y \in S$ realize two values of k. A path of adjacent cells inside S joins them, and by Fact 1.3 the value of k changes by at most one per step, so every intermediate integer occurs on the path. □

1.2 Edge perimeter, runs and directional convexity

Let $a(S)$ denote the number of unordered pairs of adjacent cells of S. The *edge perimeter* of S is

$$p(S) = 6|S| - 2a(S), \tag{1}$$

in words: six times the number of cells, minus twice the number of adjacent pairs. Equivalently, $p(S)$ counts the cell edges of S that are not shared with another cell of S; we call these the *boundary edges*. The *edge semiperimeter* is

$$\sigma(K) := \tfrac{1}{2}p(K)$$

for any finite set of cells K; in words, half the edge perimeter. That this is an integer is part of Lemma 1.6 below.

A line Λ is *occupied* by S when $S \cap \Lambda \neq \emptyset$; we call $S \cap \Lambda$ the *trace* of S on Λ. Cells of a line are linearly ordered (for a line of constant q by the value of r, for one of constant r by the value of q, for one of constant s by the value of q), and consecutive cells in that order are adjacent. A *run* of S on Λ is a maximal block of consecutive cells of the trace. Let

$$R(S) = \text{the total number of runs of } S \text{ over all lines}, \qquad \lambda(S) = \text{the number of occupied lines}.$$

Definition 1.5. A polyhex S is *directionally convex* when the trace of S on every occupied line is a single run. In words: along each of the three directions, S never leaves a line and comes back to it.

This is a lattice notion and not Euclidean convexity; it is the notion called *layer convexity* in parts of the literature on lattice animals.

Lemma 1.6. *For every finite set of cells S,*

$$p(S) = 2R(S) \;\geq\; 2\lambda(S), \tag{2}$$

in words: the edge perimeter equals twice the total number of runs, and the total number of runs is at least the number of occupied lines. In particular $p(S)$ is even and $\sigma(S) = R(S)$. For a polyhex, equality $R(S) = \lambda(S)$ holds if and only if S is directionally convex.

Proof. Let $e = \{x, y\}$ be a boundary edge, with $x \in S$ and $y \notin S$ the two cells it separates. By Fact 1.1 the cells x and y agree in exactly one key, hence they are consecutive cells of exactly one line Λ, and x is an end of the run of S on Λ containing it. Conversely a run has exactly two ends, and the cell immediately beyond each end lies outside S and determines a boundary edge; for a run consisting of a single cell the two ends are the two opposite sides of that cell, so they give two distinct boundary edges. This is a bijection between boundary edges and ends of runs, whence $p(S) = 2R(S)$.

Every occupied line carries at least one run, so $R(S) \geq \lambda(S)$, with equality exactly when every occupied line carries exactly one run, which is Definition 1.5. □

Remark 1.7 (edge perimeter versus site perimeter). The *site perimeter* of S is the number of cells outside S adjacent to at least one cell of S. It is a different functional from (1), and the two have different minimizers in general; results about one do not transfer to the other. Nothing in this paper depends on the site perimeter. We mention it only because part of the literature we cite in Section 2.2 is written in those terms.

1.3 Extreme support lines and traces

Let S be a polyhex. For each key, Fact 1.4 makes the occupied values an interval, so there is a smallest and a largest occupied line in each of the three families. These six lines are the *extreme lines* of S; we list them in the cyclic order

$$\Lambda_1 :\ q = q_{\min}, \quad \Lambda_2 :\ r = r_{\max}, \quad \Lambda_3 :\ s = s_{\max}, \quad \Lambda_4 :\ q = q_{\max}, \quad \Lambda_5 :\ r = r_{\min}, \quad \Lambda_6 :\ s = s_{\min},$$

so that Λ_i and Λ_{i+3} are the two extreme lines of one and the same family, indices being read in $\mathbb{Z}/6$. If a family occupies a single line then the two corresponding extreme lines coincide; all statements below are read with that degeneracy allowed.

Definition 1.8. Let S be a *directionally convex* polyhex. Its six *support traces* are the traces $S \cap \Lambda_i$, each a single run by Definition 1.5, and we write

$$r_i := |S \cap \Lambda_i| \ \geq 1, \qquad i \in \mathbb{Z}/6, \qquad \text{and} \qquad \rho(S) := \min_i r_i.$$

In words: r_i is the number of cells of S on the i-th extreme line, and $\rho(S)$ is the smallest of those six numbers.

Remark 1.9. Definition 1.8 is stated for directionally convex S only. On that domain $\rho(S)$ is at once the length of the shortest support trace and the length of the shortest extreme-line run. For a general polyhex a trace need not be a run, and we do not extend the definition.

2 Tight hulls and exact identities

2.1 Tight lattice hulls

Definition 2.1. Let $L \geq 1$ and $u, v, w \geq 0$ be integers satisfying the *admissibility* conditions

$$u + v \leq L - 1, \qquad v + w \leq L - 1, \qquad u + w \leq L - 1, \tag{3}$$

and put $A := L - u$, $B := L - v$, $C := L - w$, so that $A, B, C \geq 1$. The corresponding *lattice hexagon* is

$$H(L; u, v, w) := \{\, (q, r) :\ 0 \leq q \leq A - 1,\ \ 0 \leq r \leq B - 1,\ \ w \leq q + r \leq L - 1 \,\}.$$

In words: the triangle of side L with three corner triangles of sides u, v and w cut off. Degenerate shapes, in which one or more of the cut corners is empty or one of the three widths equals 1, are allowed.

Definition 2.2. The *tight hull* $H(S)$ of a polyhex S is the lattice hexagon determined by the extreme values of the three keys on S. Concretely, normalize by a translation so that $q_{\min} = r_{\min} = 0$ and set

$$A = 1 + q_{\max}, \quad B = 1 + r_{\max}, \quad L = 1 + s_{\max}, \quad w = s_{\min}, \quad u = L - A, \quad v = L - B.$$

Lemma 2.3 (the tight hull is admissible)**.** *For every polyhex S, the parameters L, u, v, w of Definition 2.2 satisfy $L \geq 1$, $u, v, w \geq 0$ and the three inequalities* (3)*, so $H(S) = H(L; u, v, w)$ is a lattice hexagon.*

Proof. After the normalization every cell has $q, r \geq 0$, hence $s \geq 0$, so $w \geq 0$ and $L \geq 1$; and $u = s_{\max} - q_{\max}$, $v = s_{\max} - r_{\max}$. A cell with $q = q_{\max}$ has $s \geq q_{\max}$, so $u \geq 0$; likewise $v \geq 0$. A cell with $s = s_{\max}$ has $s_{\max} = q + r \leq q_{\max} + r_{\max}$, which is $u + v \leq L - 1$. A cell with $q = 0$ has $s = r \leq r_{\max}$, so $s_{\min} \leq r_{\max}$, which is $v + w \leq L - 1$; a cell with $r = 0$ gives $s_{\min} \leq q_{\max}$, which is $u + w \leq L - 1$. □

For a lattice hexagon H we write $\Lambda_1, \ldots, \Lambda_6$ for its six extreme lines in the cyclic order of Section 1.3, namely

$$q = 0, \quad r = B - 1, \quad s = L - 1, \quad q = A - 1, \quad r = 0, \quad s = w,$$

and call them the *side lines*; we write

$$\ell_i := |H \cap \Lambda_i|$$

for the six *side lengths*, $h := \min_i \ell_i$ for the shortest side length, and $\kappa_i := H \cap \Lambda_i \cap \Lambda_{i+1}$ for the *corner* cells. Finally we set

$$\delta(H) := \ell_1 - \ell_4, \qquad Q(H) := (\ell_1 - \ell_3)^2 + (\ell_3 - \ell_5)^2 + (\ell_5 - \ell_1)^2.$$

Lemma 2.4. *Let S be a polyhex and $H = H(S)$. Then*

(i) $S \subseteq H$;

(ii) S and H occupy exactly the same lines, so $\lambda(S) = \lambda(H)$;

(iii) each of the six side lines of H meets S.

Proof. (i) is immediate from Definition 2.2, since every cell of S satisfies the six inequalities defining H. For (ii), the lines occupied by S in the q-family are, by Fact 1.4, exactly $q = 0, \ldots, A - 1$, and likewise $r = 0, \ldots, B - 1$ and $s = w, \ldots, L - 1$; these are precisely the lines occupied by H, as read off from Definition 2.1. Assertion (iii) is the case of (ii) for the six extreme values: each of them is attained by some cell of S. □

Lemma 2.4 is the only place where connectedness of S is used in this section, and it is used through Fact 1.4: for a disconnected set the occupied key values need not form an interval, and (ii) fails.

2.2 Hull identities

Lemma 2.5. *Let $H = H(L; u, v, w)$ be a lattice hexagon. Then:*

(i) the six side lengths are

$$\ell_1 = L - v - w, \quad \ell_2 = v + 1, \quad \ell_3 = L - u - v, \quad \ell_4 = u + 1, \quad \ell_5 = L - u - w, \quad \ell_6 = w + 1,$$

and each side line meets H in a single run whose two end cells are the corners κ_{i-1} and κ_i;

(ii) H is a directionally convex polyhex with

$$\lambda(H) = \sigma(H) = A + B + C = 3L - (u + v + w);$$

(iii) the six side lengths satisfy

$$\sum_{i=1}^{6} \ell_i = \lambda(H) + 3, \tag{4}$$

in words: the six side lengths of the hull add up to three more than the number of lines the hull occupies; and the three differences across opposite sides agree,

$$\ell_1 - \ell_4 = \ell_3 - \ell_6 = \ell_5 - \ell_2 = \delta(H) = L - u - v - w - 1; \tag{5}$$

(iv) with $T(k) := k(k+1)/2$,

$$|H| = T(L) - T(u) - T(v) - T(w);$$

(v) (hull defect identity)

$$\lambda(H)^2 + 3 - 12|H| \;=\; 3\,\delta(H)^2 + 2\,Q(H) \;\geq\; 0, \tag{6}$$

in words: the square of the number of occupied lines, plus three, minus twelve times the number of cells, equals three times the square of the common opposite-side difference plus twice the imbalance $Q(H)$.

Proof. (i) On the line $q = 0$ the defining inequalities reduce to $w \leq r \leq \min(B-1, L-1) = B-1$, which is a nonempty interval because $w \leq B-1$ is the condition $v + w \leq L-1$ of (3); its length is $B - w = L - v - w$. On the line $r = B-1$ they reduce to $\max(0, w-B+1) \leq q \leq \min(A-1, v)$, and (3) gives $w - B + 1 \leq 0$ and $v \leq A-1$, so the interval is $0 \leq q \leq v$, of length $v+1$. The remaining four sides are obtained in the same way, each time using exactly one of the three inequalities of (3) at each end. Each side is an interval of consecutive cells of its line, hence a run; its two end cells are the two cells where a second defining inequality becomes an equality, that is, the two corners.

(ii) The occupied lines are $q = 0, \ldots, A-1$, $r = 0, \ldots, B-1$ and $s = w, \ldots, L-1$, which number $A + B + C$; on each of them the trace is an interval of consecutive cells by the same computation as in (i), so H is directionally convex and Lemma 1.6 gives $\sigma(H) = R(H) = \lambda(H)$. Connectedness follows since each trace is a run and consecutive traces meet a common line.

(iii) Summing the six expressions of (i),

$$\sum_i \ell_i = 3L - (u + v + w) + 3 = \lambda(H) + 3.$$

Using the side-length formulas of (i), the three opposite-side differences $\ell_1 - \ell_4$, $\ell_3 - \ell_6$, $\ell_5 - \ell_2$ of (5) are $(L - v - w) - (u + 1)$, $(L - u - v) - (w + 1)$ and $(L - u - w) - (v + 1)$, all equal to $L - u - v - w - 1$.

(iv) The triangle $\{q, r \geq 0,\ q+r \leq L-1\}$ has $T(L)$ cells. The three sets removed in Definition 2.1 are the corner triangles $q \geq A$, $r \geq B$ and $q + r \leq w - 1$, of $T(u)$, $T(v)$ and $T(w)$ cells respectively, and (3) makes them pairwise disjoint: a cell with $q \geq A$ and $r \geq B$ would have $q + r \geq A + B = 2L - u - v \geq L + 1$, which is impossible, and a cell with $q \geq A$ and $q + r \leq w - 1$ would give $r \leq w - 1 - A < 0$, and symmetrically.

(v) Put $U := u + v + w$ and $P := u^2 + v^2 + w^2$. By (ii) and (iv),

$$\lambda(H)^2 + 3 - 12|H| = (3L - U)^2 + 3 - 6L(L+1) + 6P + 6U = 3L^2 - 6LU - 6L + U^2 + 6P + 6U + 3.$$

On the other side, $\delta(H) = L - U - 1$ and, by (i), $Q(H) = (u-w)^2 + (w-v)^2 + (v-u)^2 = 3P - U^2$, so

$$3\delta(H)^2 + 2Q(H) = 3(L - U - 1)^2 + 2(3P - U^2)$$

expands to the same expression. In words: both sides equal three times L^2, minus six times LU, minus six times L, plus U^2, plus six times P, plus six times U, plus three. Nonnegativity is clear from the right-hand side. □

Remark 2.6 (attribution). An equivalent defect identity is obtained from Fülep and Sieben's polyiamond identity [2, Lemma 3.3] through the triangular–hexagonal duality developed in their Section 5. In the same site-perimeter variables, [2, Proposition 5.10] yields exactly the capacity function $M(t)$ of Section 3.1 after the substitution $p = t + 3$; the value $p = 7$ excluded there corresponds to the value $t = 4$ excluded in Remark 3.5 below. We nevertheless give the derivation here in edge-perimeter variables to keep the argument self-contained, and we do not identify the two settings: [2] is about the site perimeter (Remark 1.7).

3 Capacity and minimum edge perimeter

3.1 Extremal capacity

We first record the two operations on directionally convex polyhexes that the capacity computation needs: removing a whole support trace, and putting it back one cell at a time.

Lemma 3.1 (peeling). *Let K be a directionally convex polyhex, let Λ be one of its six extreme lines, and let $K' := K \setminus \Lambda$. If $K' \neq \emptyset$, then*

(i) K' is a directionally convex polyhex with $\sigma(K') \leq \sigma(K) - 1$;

(ii) the cells of $K \cap \Lambda$ can be ordered $c_1, \dots, c_k$ so that for every j the set $K' \cup \{c_1, \dots, c_j\}$ is a directionally convex polyhex of edge semiperimeter at most $\sigma(K)$.

Proof. Say Λ is the line $q = q_{\min}$; the other five cases are the same after permuting the roles of the keys and of the two senses.

(i) For each occupied value k of q let K_k be the trace of K on $q = k$; it is a run. If K is connected, then for consecutive values $k, k+1$ a path inside K joining a cell of K_k to a cell of K_{k+1} must, by Fact 1.3, contain a step from a cell with $q = k$ to a cell with $q = k+1$, so K_k and K_{k+1} are adjacent. Hence K', the union of the K_k with $k \geq q_{\min} + 1$, is connected. Its traces in the q-family are traces of K, hence runs. Now fix a line of the r-family, $r = k$; its cells are (q, k), so the trace of K on it is a run if and only if its q-values form an interval, and by Fact 1.2 it meets Λ in at most the one cell $(q_{\min}, k)$. The trace of K' on this line is the trace of K with that cell, if present, removed. Since $q_{\min}$ is the least q-value occurring anywhere in K, it is the least q-value of the interval, so removing it leaves an interval, possibly empty. A line of the s-family, $s = k$, has cells $(q, k - q)$, again parametrized by q, and the same sentence applies verbatim. So all traces of K' are runs and K' is directionally convex. Finally Λ is occupied by K and not by K', so $\lambda(K') \leq \lambda(K) - 1$, and by Lemma 1.6 $\sigma(K') = \lambda(K') \leq \lambda(K) - 1 = \sigma(K) - 1$.

(ii) Since K is connected there is a cell $c_1 \in K \cap \Lambda$ adjacent to K'. Order the remaining cells of the run $K \cap \Lambda$ so that every initial segment $\{c_1, \dots, c_j\}$ is a block of consecutive cells of that run containing c_1. Write $K_j := K' \cup \{c_1, \dots, c_j\}$. Then K_j is connected, because c_1 is adjacent to K' and each later c_j is adjacent to a cell already inserted. Its trace on Λ is a block, hence a run; its trace on any other line of the q-family is a trace of K', hence a run. On a line of the r- or s-family, parametrized by q as in (i), the trace of K_j is the trace of K' there together with at most one inserted cell, which lies on Λ and so has $q = q_{\min}$. The trace of K on that line is a run containing this cell and, by (i), the trace of K' is that run with its least element $q_{\min}$ removed;

putting the cell back restores the run of K. Hence every trace of K_j is a run, so K_j is directionally convex, and every line it occupies is occupied by K, whence $\sigma(K_j) = \lambda(K_j) \le \lambda(K) = \sigma(K)$. □

Corollary 3.2. *Let K be a directionally convex polyhex. For every j with $1 \le j \le |K|$ there is a directionally convex polyhex $K_j \subseteq K$ with $|K_j| = j$ and $\sigma(K_j) \le \sigma(K)$.*

Proof. Strong induction on $|K|$. For $|K| = 1$ there is nothing to prove. Let $|K| \ge 2$. Some extreme line Λ does not contain all of K: otherwise K would be contained in two extreme lines belonging to different families, which by Fact 1.2 meet in a single cell, contradicting $|K| \ge 2$. Apply Lemma 3.1 to that Λ. The sets K_j of part (ii) supply all sizes between $|K'|$ and $|K|$; the sizes below $|K'|$ are supplied by the inductive hypothesis applied to K', whose edge semiperimeter is at most $\sigma(K)$. □

Definition 3.3. For an integer $t \ge 0$ put

$$M(t) := \left\lfloor \frac{t^2 + 3}{12} \right\rfloor,$$

in words: the integer part of one twelfth of the square of t plus three. Thus $M(0) = M(1) = M(2) = 0$ and $M(3) = 1$. The function M is nondecreasing.

Theorem 3.4. *Let $N \ge 1$.*

(a) Every polyhex S satisfies $|S| \le M(\sigma(S))$; in words, a configuration of edge semiperimeter t cannot have more than $M(t)$ cells.

(b) For every $t \ge 3$ there is a directionally convex polyhex K with $|K| = M(t)$ and $\sigma(K) \le t$.

(c) The minimum edge perimeter of an N-cell polyhex is

$$p_{\min}(N) = 2g(N), \qquad g(N) = \left\lceil \sqrt{12N - 3} \right\rceil,$$

in words: the smallest edge semiperimeter that can hold N cells is the least integer at least as large as the square root of twelve times N minus three, and the minimum edge perimeter is twice that number.

(d) Every N-cell minimizer S is directionally convex and satisfies $\sigma(S) = \lambda(S) = g(N)$.

(e) Writing $\mathrm{exc}(S) := p(S) - p_{\min}(|S|)$, a polyhex S satisfies $\mathrm{exc}(S) \le 2$ if and only if $|S| > M(\sigma(S) - 2)$.

The directional convexity in (d) is a known result: Davoli, Piovano and Stefanelli [1, Proposition 3.3] prove that every minimizer of the edge-isoperimetric problem on the triangular lattice is 3-convex, which is directional convexity in the present terminology; the proof below is the self-contained one of the main paper, via the hull identities.

Proof. (a) Let $H = H(S)$. By Lemma 2.4 and Lemma 2.5(v),

$$12|S| \le 12|H| = \lambda(H)^2 + 3 - \big(3\delta(H)^2 + 2Q(H)\big) \le \lambda(S)^2 + 3 \le \sigma(S)^2 + 3,$$

the last step by Lemma 1.6. Since $|S|$ is an integer, $|S| \le M(\sigma(S))$.

(b) For $t \equiv 3 \pmod 6$, say $t = 6m + 3$, take $K = H(3m + 1; m, m, m)$; for $t = 6m$ take $H(3m - 1; m - 1, m - 1, m - 1)$; for $t = 6m + 1$ take $H(3m; m, m, m - 1)$; for $t = 6m + 2$ take $H(3m + 1; m, m, m + 1)$; for $t = 6m + 4$ take $H(3m + 1; m, m, m - 1)$; for $t = 6m + 5$ take

$H(3m+2; m, m, m+1)$. In each case (3) holds for the indicated range of m, Lemma 2.5(ii) gives $\sigma(K) = t$, and Lemma 2.5(iv),(v) give $12|K| = t^2 + 3 - D$ with $D \in \{0, 3, 4, 7\}$ according to the residue of t, so $|K| = M(t)$. The single case not covered by an admissible triple is $t = 4$; there $M(4) = 1$ and a single cell has edge semiperimeter $3 \le 4$.

(c) Put $t_N := \min\{t \ge 3 : \ M(t) \ge N\}$. By (a) every N-cell polyhex has $M(\sigma(S)) \ge N$, hence $\sigma(S) \ge t_N$. Conversely (b) produces a directionally convex K with $|K| = M(t_N) \ge N$ and $\sigma(K) \le t_N$, and Corollary 3.2 extracts from it an N-cell polyhex of edge semiperimeter at most t_N. Therefore $p_{\min}(N) = 2t_N$. Finally, for integers t,

$$M(t) \ge N \iff \frac{t^2+3}{12} \ge N \iff t \ge \sqrt{12N-3},$$

and $\sqrt{12N-3} \ge 3$ for $N \ge 1$, so $t_N = g(N)$.

(d) By (c), $\sigma(S) = g(N)$. Let $H = H(S)$. Applying (a) to H, which is directionally convex with $\sigma(H) = \lambda(H)$, gives $M(\lambda(H)) \ge |H| \ge N$ and therefore $\lambda(H) \ge t_N = g(N)$. Combining with Lemmas 1.6 and 2.4,

$$g(N) = \sigma(S) = R(S) \ \ge \ \lambda(S) = \lambda(H) \ \ge \ g(N),$$

so $R(S) = \lambda(S)$ and S is directionally convex.

(e) Write $t = \sigma(S)$ and $k = |S|$; by (c), $\operatorname{exc}(S) = 2\big(t - g(k)\big)$. Since both sides are integers, $\operatorname{exc}(S) > 2$ is equivalent to $g(k) \le t - 2$. If $t - 2 \ge 3$, then by the definition of g as t_k and the monotonicity of M this is equivalent to $M(t-2) \ge k$. If $t - 2 < 3$, then $t \in \{3, 4\}$ because a polyhex has $t \ge 3$; here $M(t-2) = 0 < k$ and $g(k) \ge 3 > t - 2$, so both statements are false. In both cases $\operatorname{exc}(S) > 2$ is equivalent to $M(t-2) \ge k$, which is the assertion. □

Remark 3.5. Parts (a) and (b) together say exactly that

$$M(t) = \max\{\, |K| : \ K \text{ a polyhex with } \sigma(K) \le t \,\},$$

in words: $M(t)$ is the largest number of cells that fit within an edge-semiperimeter budget of t. The budget form is the one the rest of the paper uses, and the inequality in (b) is not a weakening: for every $t \ge 3$ other than $t = 4$ the hexagons listed in the proof do satisfy $\sigma(K) = t$, and the value $t = 4$ is not the edge semiperimeter of any polyhex at all. Indeed a polyhex K with $\sigma(K) = 4$ would have $|K| \le M(4) = 1$ by (a), while the one-cell polyhex has edge semiperimeter 3.

3.2 The plateau bridge

The function $N \mapsto g(N)$ is nondecreasing and takes each of its values on a block of consecutive integers. For γ in the range of g let

$$N_b(\gamma) := \min\{\, N \ge 1 : \ g(N) = \gamma \,\}$$

be the first cell count at which the minimum edge semiperimeter equals γ.

Lemma 3.6. *For every γ in the range of g,*

$$N_b(\gamma) = M(\gamma - 1) + 1, \tag{7}$$

in words: the plateau on which the minimum edge semiperimeter equals γ begins one cell after the capacity of edge semiperimeter $\gamma - 1$ has been exhausted.

Proof. For integers $N \ge 1$ and $t \ge 3$ we have, as in the proof of Theorem 3.4(c), $g(N) \le t$ if and only if $N \le M(t)$. Hence $\{N : \ g(N) \le \gamma - 1\} = \{1, \dots, M(\gamma - 1)\}$, so the least N with $g(N) \ge \gamma$ is $M(\gamma - 1) + 1$; as γ lies in the range of g, that least N satisfies $g(N) = \gamma$. □

4 Near-extremal configurations and missing cells

Definition 4.1. A directionally convex polyhex S is *near-extremal* when

$$|S| \;\geq\; M\big(\sigma(S)-1\big),$$

in words: S has at least as many cells as the largest configuration whose edge semiperimeter budget is one unit smaller. We call the family of these configurations the *near-extremal class*.

Lemma 4.2. *Every edge-perimeter minimizer is near-extremal.*

Proof. Let $|S| = N$. By Theorem 3.4(d), S is directionally convex with $\sigma(S) = g(N)$. By the definition of g as the least t with $M(t) \geq N$ we have $M(g(N)-1) < N$, so $|S| = N > M(\sigma(S) - 1)$. □

The class is strictly larger than the class of minimizers: it contains configurations of excess 2 as well as of excess 0. The main paper shows that peeling a shortest extreme line from a minimizer always produces a minimizer, so the induction there can be run inside the minimizers; this note shows that the same induction closes on the wider class, at the price of a stronger peeling estimate (Lemma 5.1).

Notation for Sections 4–6

Throughout, S is a directionally convex polyhex with

$$n = |S|, \qquad \sigma = \sigma(S) = \lambda(S), \qquad H = H(S),$$

so that $\sigma(H) = \sigma$ by Lemma 2.4(ii). We use the *defect*

$$D(S) = \sigma^2 + 3 - 12n, \tag{8}$$

in words: the square of the edge semiperimeter, plus three, minus twelve times the number of cells. The cells of $H \setminus S$ are called *missing*, their number is $\mu = |H| - n$, and (6) gives

$$D(S) = D(H) + 12\mu, \tag{9}$$

in words: the defect of S is the defect of its hull plus twelve for each missing cell. For $i \in \mathbb{Z}/6$ we write, as in Definition 1.8 and Section 2.1,

$$r_i = |S \cap \Lambda_i| \;\geq 1, \qquad m_i = \ell_i - r_i \;\geq 0, \qquad \beta = \sum_i m_i, \qquad \rho = \rho(S) = \min_i r_i,$$

so m_i counts the missing cells on the i-th side line and β counts, with multiplicity, the missing cells lying on side lines.

Lemma 4.3 (missing cells)**.** *Let S be a directionally convex polyhex and $H = H(S)$.*

(a) If $m_i \geq 1$ then κ_{i-1} or κ_i is missing; consequently $m_{i-1} \geq 1$ or $m_{i+1} \geq 1$.

(b) For every missing cell z and every line Λ through z, one of the two end cells of $H \cap \Lambda$ is missing, and every end cell of a trace of H lies on a side line.

(c) If $\beta = 0$ then $S = H$.

(d) If some missing cell lies on no side line, then at least three distinct missing cells lie on side lines.

Proof. (a) By Lemma 2.5(i) the trace $H \cap \Lambda_i$ is a run with end cells κ_{i-1} and κ_i, and $S \cap \Lambda_i$ is a nonempty run inside it. The complement of one run inside another is the union of an initial and a final segment, so if it is nonempty it contains an end cell of $H \cap \Lambda_i$. Since $\kappa_{i-1} \in \Lambda_{i-1}$ and $\kappa_i \in \Lambda_{i+1}$, the second assertion follows.

(b) Let z be missing and Λ a line through z. Then Λ is occupied by H, hence by S as well by Lemma 2.4(ii). Now $H \cap \Lambda$ is a run containing z, and $S \cap \Lambda$ is a run not containing z; therefore z lies in one of the two segments of $(H \cap \Lambda) \setminus (S \cap \Lambda)$, and every cell of that segment, in particular the end cell of $H \cap \Lambda$ on that side, is missing. For the second assertion, let e be an end cell of a trace $H \cap \Lambda$. The next cell of Λ beyond e is not in H, so it violates one of the six inequalities of Definition 2.1; since one adjacency step changes each key by at most 1 (Fact 1.3), e satisfies that same inequality with equality, which says exactly that e lies on the corresponding side line.

(c) If no missing cell lies on a side line then by (b) there is no missing cell at all.

(d) Let z be missing and on no side line, and apply (b) to the three lines through z, obtaining missing end cells e_1, e_2, e_3 on side lines. Each $e_j \neq z$ because z is on no side line. If $e_j = e_k$ for $j \neq k$, then z and this common cell would both lie on a line of the j-th family and on a line of the k-th family, and two lines of different families meet in exactly one cell (Fact 1.2), forcing $z = e_j$. So e_1, e_2, e_3 are distinct. □

5 The key peeling lemma

Lemma 5.1 (key lemma)**.** *Let S be near-extremal with $n = |S| \geq 2$. Then*

$$n - \rho \;\geq\; M(\sigma - 2), \tag{10}$$

in words: deleting the shortest support trace still leaves at least as many cells as fit within an edge semiperimeter budget two units smaller. Moreover there is an index i with $r_i = \rho$ and $S \setminus \Lambda_i \neq \emptyset$.

Proof. We prove the last assertion first. Suppose that $S \subseteq \Lambda_i$ for every i with $r_i = \rho$. For such an i we get $n = |S \cap \Lambda_i| = r_i = \rho$, hence $n = \rho \leq r_j \leq n$ and therefore $r_j = n$, that is $S \subseteq \Lambda_j$, for every j. Taking two side lines of different families and applying Fact 1.2 gives $n \leq 1$, contradicting $n \geq 2$.

For (10) we argue by contradiction, in six steps. Assume $n - \rho < M(\sigma - 2)$; as all quantities are integers,

$$n - \rho + 1 \;\leq\; M(\sigma - 2). \tag{11}$$

Step 1. *For every i,*

$$12\, r_i \;\geq\; 4\sigma + 8 - D(S). \tag{12}$$

Indeed $r_i \geq \rho$, so by (11) and Definition 3.3,

$$n - r_i + 1 \;\leq\; n - \rho + 1 \;\leq\; M(\sigma - 2) \;\leq\; \frac{(\sigma - 2)^2 + 3}{12}.$$

Multiplying by 12 and using (8) gives $12n - 12r_i + 12 \leq \sigma^2 - 4\sigma + 7$, which rearranges to (12). In words: twelve times any support-trace length is at least four times the edge semiperimeter, plus eight, minus the defect.

Step 2. $D(S) \le 2\sigma+10$. Near-extremality gives $n \ge M(\sigma-1) > \frac{(\sigma-1)^2+3}{12} - 1$, so $12n > \sigma^2 - 2\sigma - 8$ and hence $D(S) < 2\sigma + 11$. Both sides being integers, $D(S) \le 2\sigma + 10$.

Step 3. *Put $z_i := \ell_i - \rho$ and $W := \sum_i z_i$. Then $z_i \ge m_i \ge 0$, $W \le 4$, and $D(S) \ge 4W + 8$.* For the first assertion,

$$z_i = \ell_i - \rho = (r_i - \rho) + (\ell_i - r_i) = (r_i - \rho) + m_i \;\ge\; m_i \;\ge\; 0,$$

since $r_i \ge \rho$; in words, the slack of a hull side over the shortest support trace is at least the number of missing cells on that side. By (4), $\sum_i \ell_i = \sigma + 3$, so

$$6\rho + W = \sigma + 3, \qquad \text{equivalently} \qquad 12\rho = 2\sigma + 6 - 2W. \tag{13}$$

Applying (12) to an index with $r_i = \rho$ and substituting (13),

$$2\sigma + 6 - 2W \;\ge\; 4\sigma + 8 - D(S), \qquad \text{that is} \qquad D(S) \;\ge\; 2\sigma + 2 + 2W.$$

Together with Step 2 this forces $W \le 4$. Support traces are nonempty by Lemma 2.4(iii), so $\rho \ge 1$, and (13) gives $\sigma = 6\rho + W - 3 \ge W + 3$; substituting $2\sigma \ge 2W + 6$ into $D(S) \ge 2\sigma + 2 + 2W$ yields $D(S) \ge 4W + 8$.

Step 4. *Up to a cyclic rotation of the indices, the vector $(z_1, \dots, z_6)$ is one of*

$$\text{(i) } (0,0,0,0,0,0), \qquad \text{(ii) } (1,0,0,1,0,0), \qquad \text{(iii) } (1,0,1,0,1,0),$$

$$\text{(iv) } (2,0,0,2,0,0), \qquad \text{(v) } (1,1,0,1,1,0).$$

Since $z_i = \ell_i - \rho$, the differences of the z_i are those of the ℓ_i, so (5) gives

$$z_1 - z_4 = z_3 - z_6 = z_5 - z_2 = \delta, \qquad \delta := \delta(H).$$

If $\delta \ge 0$ then $W = 3\delta + 2(z_2 + z_4 + z_6) \ge 3\delta$, and symmetrically for $\delta \le 0$; with $W \le 4$ this gives $|\delta| \le 1$. If $|\delta| = 1$ then $3 + 2(\cdot) \le 4$ forces the bracket to vanish, leaving $W = 3$ and the vector $(1,0,1,0,1,0)$ or its rotation $(0,1,0,1,0,1)$, which is (iii). If $\delta = 0$ then $z_{i+3} = z_i$ and $W = 2(z_1 + z_2 + z_3)$, so $z_1 + z_2 + z_3 \in \{0,1,2\}$; the three possible partitions give (i), (ii) and, for the value 2, either (iv) or (v).

The rotation is harmless in Steps 5 and 6, which read the vector only through quantities that do not depend on where the pattern sits. This is because i and $i+3$ always have opposite parities, so each pair of opposite indices contains exactly one odd index; the counts of positive entries among z_1, z_3, z_5 used below are therefore the same for every rotation. (When $\delta = 0$ one also has $\ell_{i+3} = \ell_i$, so the two alternating triples of side lengths agree as multisets and $Q(H)$ may be read off from either.)

Step 5. *Patterns* (i)–(iv) *are impossible.* In each of them no two cyclically consecutive z_i are positive, hence by Step 3 no two cyclically consecutive m_i are positive either. But Lemma 4.3(a) says that $m_i \ge 1$ forces $m_{i-1} \ge 1$ or $m_{i+1} \ge 1$. So every $m_i = 0$, that is $\beta = 0$, and Lemma 4.3(c) gives $S = H$ and $\mu = 0$. By (9) and (6), $D(S) = 3\delta^2 + 2Q(H)$, and $Q(H)$ is computed from the differences of z_1, z_3, z_5. The four cases give:

pattern		W	δ	$Q(H)$	$D(S)$	Step 3 requires
(i)	$(0,0,0,0,0,0)$	0	0	0	0	≥ 8
(ii)	$(1,0,0,1,0,0)$	2	0	2	4	≥ 16
(iii)	$(1,0,1,0,1,0)$	3	± 1	0	3	≥ 20
(iv)	$(2,0,0,2,0,0)$	4	0	8	16	≥ 24

The values of $Q(H)$ are obtained as follows. In (ii) the two positive entries occupy one pair of opposite indices, which contains exactly one odd index, so (z_1, z_3, z_5) is a permutation of $(1, 0, 0)$ and $Q(H) = 1 + 0 + 1 = 2$; in (iv) the same argument with the value 2 gives $Q(H) = 4 + 0 + 4 = 8$; in (iii) the triple (z_1, z_3, z_5) is $(1, 1, 1)$ or $(0, 0, 0)$ and $Q(H) = 0$. Each row contradicts the bound of Step 3.

Step 6. *Pattern* (v) *is impossible.* Here $\delta = 0$ and the positive entries are $z_j = z_{j+1} = z_{j+3} = z_{j+4} = 1$ for some j, with $z_{j+2} = z_{j+5} = 0$. Among z_1, z_3, z_5 exactly two equal 1: the indices j and $j+3$ have opposite parities and contribute exactly one, and so do $j+1$ and $j+4$. Hence (z_1, z_3, z_5) is a permutation of $(1, 1, 0)$, $Q(H) = 0 + 1 + 1 = 2$, and $D(H) = 3 \cdot 0 + 2 \cdot 2 = 4$.

By Step 3, $m_{j+2} = m_{j+5} = 0$ and every other m_i is at most 1. Consider Λ_j: if $m_j \geq 1$ then by Lemma 4.3(a) one of κ_{j-1}, κ_j is missing, and $\kappa_{j-1} \in \Lambda_{j-1} = \Lambda_{j+5}$, whose m vanishes; so the missing cell is κ_j. On Λ_{j+1} the two candidates are κ_j and $\kappa_{j+1} \in \Lambda_{j+2}$, and the latter is excluded, so again only κ_j. On Λ_{j+3} the candidates are $\kappa_{j+2} \in \Lambda_{j+2}$, excluded, and κ_{j+3}; on Λ_{j+4} they are κ_{j+3} and $\kappa_{j+4} \in \Lambda_{j+5}$, excluded. So the only cells on side lines that can be missing are κ_j and κ_{j+3}: at most two of them. By Lemma 4.3(d) a missing cell lying on no side line would force three distinct missing cells on side lines, so there is none, and

$$\mu \leq 2, \qquad D(S) = D(H) + 12\mu = 4 + 12\mu \in \{4, 16, 28\}.$$

On the other hand $W = 4$, so Step 3 gives $D(S) \geq 2\sigma + 10$ while Step 2 gives $D(S) \leq 2\sigma + 10$; hence $D(S) = 2\sigma + 10$. Since also $D(S) \geq 4W + 8 = 24$, the only possibility is $D(S) = 28$, whence $\sigma = 9$ and $\mu = 2$. But (13) with $W = 4$ gives $\sigma = 6\rho + 1$, so $\sigma \equiv 1 \pmod 6$, while $9 \equiv 3 \pmod 6$. This contradiction completes the proof. □

6 The primary proof

Theorem 6.1. *Let S be a near-extremal polyhex with $n = |S|$. Then there are polyhexes*

$$S_1 \subset S_2 \subset \cdots \subset S_n = S, \qquad |S_k| = k,$$

each directionally convex, with $\mathrm{exc}(S_k) \leq 2$ *for every k.*

Proof. Strong induction on n. For $n = 1$ the single cell has $p = 6 = p_{\min}(1)$ and exc $= 0$.

Let $n \geq 2$. By Lemma 5.1 there is an index i with $r_i = \rho$ and $S' := S \setminus \Lambda_i \neq \emptyset$. By Lemma 3.1(i), S' is a directionally convex polyhex with $\sigma(S') \leq \sigma - 1$, and $|S'| = n - \rho$. Using (10) and the monotonicity of M,

$$|S'| = n - \rho \;\geq\; M(\sigma - 2) \;\geq\; M\big(\sigma(S') - 1\big),$$

so S' is again near-extremal. In words: peeling the shortest support trace lowers the budget by at least one and removes at most what that budget can hold, so the configuration stays in the class. By the inductive hypothesis there is a chain $S_1 \subset \cdots \subset S_{n-\rho} = S'$ of directionally convex polyhexes with $|S_k| = k$ and $\mathrm{exc}(S_k) \leq 2$.

By Lemma 3.1(ii) the cells of $S \cap \Lambda_i$ can be ordered $c_1, \ldots, c_\rho$ so that each

$$T_j := S' \cup \{c_1, \ldots, c_j\} \qquad (1 \leq j \leq \rho)$$

is a directionally convex polyhex with $\sigma(T_j) \leq \sigma$. Put $S_{n-\rho+j} := T_j$; then $T_\rho = S$ and the sizes increase by one at each step. Finally, $|T_j| \geq n - \rho + 1 > M(\sigma - 2) \geq M\big(\sigma(T_j) - 2\big)$ by (10) and the monotonicity of M, so Theorem 3.4(e) gives $\mathrm{exc}(T_j) \leq 2$. □

Corollary 6.2. *Every edge-perimeter minimizer admits a one-cell growth sequence by adjacent cells with excess at most two at every stage.*

Proof. By Lemma 4.2 the minimizer is near-extremal; apply Theorem 6.1. Adjacency of each added cell follows from connectedness of each stage. □

Remark 6.3 (the procedure). The proof is constructive and the construction is short to state: repeatedly delete the whole shortest support trace, and read the resulting sequence backwards, restoring each deleted trace one cell at a time from a cell adjacent to what remains. Each configuration along the way is directionally convex and has excess 0 or 2.

Remark 6.4 (relation to the main paper). For minimizers the main paper obtains the strict inequality $n-\rho \geq M(\sigma-2)+1$, hence a minimizer after peeling, from a residue computation and a rigidity statement at two exceptional plateau starts. Lemma 5.1 above obtains the weaker $n-\rho \geq M(\sigma-2)$ on the wider near-extremal class by a finite case analysis of the hull; that is exactly what is needed for the wider class to be closed under the peeling step, and Theorem 6.1 is correspondingly stronger in scope than the growth theorem of the main paper, though not needed for it.

References

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